INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

GRADER, HEAVY, ROAD, MOTORIZED, DED,

COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a commercial model, heavy-duty, pneumatic-tired, six-wheel, four-wheel drive, front-wheel steer, diesel-engine-driven, articulated frame motor grader.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook form a part of this specification to the extent specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

AA-1108	- Extinguisher, Tire, Vaporizing Liquid.
VV-F-800	- Fuel Oil, Diesel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATIN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-V-173 - Varnish, Moisture- and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-G-762 - Graders, Road, Packaging of.
MIL-C-1283 - Can, Gasoline, Military, 5-Gallon.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-M-7866 - Molybdenum Disulfide, Technical,Lubrication Grade.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-I-14511 - Insulation Sheet, Cellular, Plastic: Thermal.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354,Metric.
MIL-S-40626 - Sign Kit, Vehicle Class.
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited,Heavy Duty, Single Package.
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-B-46176 - Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative,Metric.
MIL-T-62314 - Test Equipment (Simplified) for Internal Combustion Engines-Reprogrammable (STE/ICE-R) Test Set.
MIL-T-83133 - Turbine Fuel, Aviation Kerosene Type, Grade JP-8.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-838 - Lubrication of Military Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.
MS35424 - Light, Marker, Clearance - Blackout, 24 Volt.
MS51318 - Headlight: Blackout, 24 Volt,Waterproof.
MS51330 - Stop Light - Taillight, Vehicular-24Volt, Blackout Tail, Blackout Stop.
MS52131 - Connector, Plug, Electrical Intervehicular Power Cable.
MS53052 - Bracket Assembly, Liquid Container, Five Gallon.

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HANDBOOKS

MILITARY

MIL-HDBK-113 - Guide for the Selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

MILITARY TRAFFIC MANAGEMENT COMMAND (MTMC/TEA)

Phamphlet 70-1 - Transportability for Better Strategic Mobility.

(Application for copies should be addressed to the military Traffic Management Command, Transportation Engineering Agency (MTMC/TEA), ATTN: MTT-TR, P.O. Box 6276, Newport News, VA 23606-0276.)

DEPARTMENT OF LABOR

OSHA 29 CFR 1926 - Occupation Safety and Health Administration

DEPARTMENT OF TRANSPORTATION (DOT)

DOT Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

DRAWINGS

TACOM

12258941 - Connector, Receptacle.
12258955 - Vehicle Test Card.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Tank Automotive Command, ATTN: AMSTA-MD, Warren, MI 48397-5000.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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ASSOCIATION OF AMERICAN RAILROADS (ARR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, American Railroads Building, 50 F Street, NW, Suite 7704, Washington, D.C. 20001.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J53	- Minimum Performance Criteria for Emergency Steering of Wheeled Earthmoving Construction Machines.
SAE J209	- Instrument Face design and Location for Construction and Industrial Equipment.
SAE J321	- Minimum Performance Criteria for Falling Object Protective Structure (FOPS).
SAE J276	- Steering Frame Lock for Articulated Loaders and Tractors.
SAE J386	- Operator Restraint Systems for Off-Road Work Machines.
SAE J514	- Hydraulic Tube Fittings.
SAE J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
SAE J614	- Engine and Transmission Dipstick Marking.
SAE J678	- Speedometers and Tachometers-Automotive.
SAE J872	- Drawbar Test Procedure for Construction, Forestry, and Industrial Machines.
SAE J994	- Alarm-Backup-Electric-Performance, Test, and Application.
SAE J1040	- Performance Criteria for Rollover Protective Structures (ROPS) for Construction, Earthmoving, Forestry, and Mining Machines.
SAE J1152	- Braking Performance-Rubber-Tired Construction Machines.
SAE J1349	- Engine Power Test Code-Spark Ignition and Diesel.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION (TRA)

Tire and Rim Association Yearbook, with Military Supplement.

(Application for copies should be addressed to the Tire and Rim Association, 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The grader shall be the latest model of the standard product of the contractor and shall have been marketed and in commercial field use in significant numbers for at least 1 year prior to the invitation for bids or request for proposal. Product improvements incorporated in the commercial production units during this 1-year period are acceptable. When specific requirements are not stated herein, all items listed as standard equipment in the contractor's published specification brochures and catalogs, or normally furnished to commercial customers as standard equipment, shall be furnished. The grader shall be equipped with optional equipment as specified herein. Optional equipment is defined as equipment not standard with the grader but which has been furnished to the commercial customer, such as special features or allied equipment. The grader shall be equipped with all components necessary to enable it to function reliably and efficiently in sustained operation. The grader shall conform to all Federal laws and regulations applicable to the manufacturer governing safety, and pollution, which are in effect for this type of equipment at the time of manufacture. The grader shall be heavy-duty, pneumatic-tired, diesel-engine-driven, six-wheel, four-wheel drive, with front wheel steer and articulated frame steer. The grader shall be equipped with a full power shift transmission and a scarifier attachment. The grader shall have leaning front wheels capable of leaning not less than 15 degrees right and left. All movement of the blade, scarifier, and leaning wheel mechanisms shall be controlled through hydraulic valves. The operating weight of the grader shall be 29,500 to 33,500 pounds. Weight shall be determined with the grader in a fully serviced condition, including fuel, oil, water, scarifier, rollover protective structure (ROPS), falling object protective structure (FOPS), and cab with standard accessories. Weight shall not include the operator or such optional items as hydroflation or other tire ballast; cast hubs, bolt-on wheel weights, or other bolted-on "dead" weight; or oversized equipment for the sole purpose of increasing operating weight. The grader shall have a lock-unlock drive train differential. The grader shall be designed to permit true "sit-down" operation with operator seated and seat belts properly fastened.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Production graders. All graders furnished under this specification shall be physically and mechanically identical to the first article grader approved by the Government, except that product improvements incorporated in the contractor's standard commercial product lines are permitted at no cost to the Government subject to the following:

a. Physical and performance characteristics have not been degraded.
b. New components involving Government selected repair parts are directly interchangeable with the superseded component or part.
c. The Government is notified of all changes to any item on the recommended parts list where fit and function is effected a minimum of 45 days prior to the production date of the change. When the changes and alterations have been incorporated in the production grader, the Government shall be furnished, by serial number, the dates production becomes effective.

3.4 Materials. Materials not specified herein shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.4.1 Material deterioration prevention and control. The grader shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the grader may be exposed.

3.4.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.4.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the grader may be newly fabricated from recovered materials to the maximum extent practicable, provided the grader produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the grader.

3.5 Cab. The grader shall be equipped with a removable, bolted-down, fully enclosed cab with an integral rollover protective structure and falling object protective structure (ROPS and FOPS). The cab shall be removable and replaceable without cutting or welding. The ROPS shall conform to SAE J1040 and the FOPS shall conform to SAE J231. The ropes shall also comply with applicable OSHA 29 CFR 1926. A seat belt conforming to SAE J386 shall be properly installed for use by the operator. The cab shall not obstruct the view of the grader and blade operation while operator is seated. The cab shall be equipped with two doors, one on each side of the cab, and each door shall be provided with a lock or latch to secure the door in an open position. A personnel heater, front and rear defroster fans, front and rear windshield wipers, inside rear view mirror, and outside rear view mirrors mounted on the right- and left-hand side of the cab shall be furnished. Quick-disconnect fittings (for example, hydraulic or electric) shall be provided as applicable to facilitate cab removal. A slinging eye conforming to MIL-STD-209 shall be provided to facilitate removal of the structure from the grader.

3.6 Seat. The contractor's standard commercial seat shall be furnished. The seat shall be located so when the operator is seated with seat belt fastened and properly adjusted, he has convenient access to operating controls as well as visibility of the work zone and all instruments and indicators mounted on the instrument panel.

3.7 Horn. A conventional operator-controlled horn shall be furnished.

3.8 Back-up alarm. A back-up alarm shall be provided on the grader and shall comply with SAE J994, type C. The back-up alarm signal shall be audible above, and distinguishable from the surrounding noise level.

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3.9 Articulation position indicator. The grader shall be equipped with an articulation position indicator.

3.10 Diesel engine. The diesel engine shall meet the requirements of MIL-STD-1410 for application in a class II end item. The engine shall be the heavy-duty diesel furnished commercially, including all systems, components, accessories, and auxiliaries. The applied net horsepower of the engine at the flywheel when measured in accordance with SAE J816 shall be not less than 135 horsepower. The engine shall be equipped with a muffler. The exhaust stack shall be equipped with a rain cap or shall be fabricated and mounted to prevent the entry of rain or snow. The engine shall be the same as that used in the manufacturer's standard commercial grader. An air-cleaner restriction gage shall be furnished. The engine lubricating and fuel supply systems shall maintain operating pressure on 35-percent side slopes and 60-percent longitudinal slopes. The diesel engine furnished shall operate on fuel conforming to both VV-F-800, grade DF-2 and MIL-T-83133, grade JP-8.

3.10.1 Engine side panels. The grader shall be equipped with removable engine side panels with quick-disconnect latches on horizontally or vertically hinged engine compartment doors. The compartment doors shall be self-supporting in the open position.

3.10.2 Oil sampling valve. An oil sampling valve shall be provided on the engine and transmission. The valve shall be manually operated and shall have the capability of closing automatically when released. It shall be made of material resistant to corrosion so that it will not contaminate the sample. The discharge port of the valve shall be covered with a captive chained cap conforming to SAE J514. The valve shall be located in such a way as to ensure that personnel will not be exposed to danger when taking oil samples with the engine running. The oil tap shall be located so that when samples are taken, it shall be a true representation of oil that is flowing while the engine is running. Sampling valves shall be labelled crankcase or transmission, whichever is appropriate, to indicate the type of oil sampled.

3.10.3 Engine cooling system. The contractor's standard or optional cooling system meeting the environmental requirements of 3.29 shall be provided.

3.10.4 Tachometer drive. A heavy-duty key type tachometer drive shall be provided and shall have a 0.15625-inch female connection in accordance with SAE J678. The output drive shall be covered by a threaded captive-chained cap for protection while not in use.

3.11 Transmission and torque converter.

3.11.1 Transmission. The grader shall be equipped with a power shift transmission capable of being shifted under full engine power through all successive forward and reverse gear ratios without deceleration of the grader. Not less than four forward speeds and four reverse speeds shall be provided by the transmission.

3.11.2 Torque converter. Graders equipped with torque converter drives shall have gear reductions suitable to keep the torque converter working efficiently throughout the entire range of grader speeds.

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3.11.3 Tandems and differential. Tandem oscillation shall be not less than 16 degrees from the horizontal in either direction. Tandem oscillation shall be limited to prevent excessive rotation when the grader is being lifted, using the lifting attachments. The grader shall have a manual or automatic lock-unlock drive train differential.

3.12 Fuel tank. The fuel tank capacity shall permit grader operation for a minimum of 10 hours under normal operation without refueling. Fuel capacity shall be verified during conduct of the construction task (see 4.5.2.18). The fuel tank shall be equipped with a means of draining water and sediment from the bottom of the tank. The fuel fill port shall be labeled "DIESEL FUEL ONLY" or "JP-8 FUEL ONLY," whichever is appropriate.

3.13 Steering and braking.

3.13.1 Steering. The grader shall have hydraulically boosted, mechanical or full hydraulic power steering. The steering system shall provide full control of the grader under normal operating conditions and shall provide emergency steering with a force not exceeding 50 pounds and conforming to SAE J53 without engine power. The steering system shall limit the turning radius to 31 feet.

3.13.2 Braking system. A foot-controlled, self-adjusting, sealed disc, service braking system shall be provided and shall be effective on the four driving wheels. Single drive shaft or two-wheel service brakes are not acceptable. A parking brake system and an emergency stopping system shall be provided. In event of power assist failure, the service brake shall be operable with a force not exceeding 70 pounds. The braking system shall comply with SAE J1152. When nonpetroleum based hydraulic brake fluid is used in a hydraulic braking system, it shall be of the silicone type in accordance with MIL-B-46176. Asbestos material shall not be used in the brake system for brake shoes or linings on this vehicle.

3.13.2.1 Air brake system. When an air brake system is provided, an alcohol evaporator with shatter resistant translucent container or air dryer system with automatic drain valve shall be provided.

3.14 Hydraulic system. The grader shall be equipped with the contractor's standard hydraulic system which shall provide sufficient force for operating all attachments and related components. All movements of the blade, scarifier, and leaning wheel mechanisms shall be controlled through hydraulic valves.

3.15 Wheels and tires.

3.15.1 Wheels. Commercial-standard demountable wheels or rims shall be provided. Front and rear wheels shall be interchangeable.

3.15.2 Tires. Tires shall be furnished. They shall be of the tubeless type and shall have a tread pattern suitable for grader operation. Tire size shall be 13 x 24 with not less than a 10-ply rating. All tires shall be the same size, type, and ply rating. The recommended tire pressure in accordance with Tire and Rim Manufacturing Yearbook recommendations shall be stencilled "TP (specify) LBS" in one-inch-high letters near each tire. Tire furnished to the Government shall be not more than 12 months old from date of cure of the rubber to date of acceptance by the Government.

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3.16 Electrical system. The manufacturer's standard or optional electrical system shall be provided that meets the requirements specified herein to include the blackout lighting system. A 24-volt starting system shall be furnished.

3.16.1 Lighting. The following minimum number of lights (meeting DOT Motor Carrier Safety Regulations, as applicable) with associated circuits and switches shall be provided:

a. Two headlights (cab mounted).

b. Two front working lights.

c. Combination stop and taillights.

d. Turn signals with emergency flasher.

e. Rear floodlight.

f. Instrument panel lighting.

3.16.2 Alternator. The manufacturer's high-output option alternator shall be furnished with not less than 50 amps.

3.16.3 Instruments and indicators. The following instruments and indicators shall be provided:

a. Ammeter, voltmeter or battery charging indicator light or gage.

b. Engine cooling liquid temperature.

c. Engine lubrication oil pressure.

d. Hourmeter.

e. Transmission and torque converter temperature gage.

f. Fuel level indicator (gage or dipstick).

g. Air pressure gage (when equipped with air system).

h. Low-air-pressure warning buzzer (when equipped with air system).

i. Hydraulic oil level indicator (dipstick, sight tube or gage). If a dipstick is used, the dipstick marking shall comply with SAE J6146.

j. Air cleaner restriction gage.

Additional instruments and indicators standard with the contractor's commercial practice, which contribute to the safety and reliability of the grader operation, shall be furnished. All instruments shall comply with SAE J209.

3.17 Battery. The battery or batteries shall be the contractor's standard or optional batteries for cold weather starting as specified in 3.29. If batteries

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are placed in a battery box, there shall be a warning placed on the box to warn of the presence of toxic fumes and the explosive hazards.

3.17.1 Slaving components. The grader shall be equipped with a 24-volt slave receptacle conforming to MS52131. The slave receptacle shall permit charging of the grader batteries and slave starting of the engine from an external power source and shall provide a power source for charging and slaving other equipment. The slave receptacle shall be installed on the exterior of the grader near the battery enclosure and shall be accessible to personnel standing on the ground. The voltage potential of the slave receptacle shall be identified on a plate furnished adjacent to the receptacle. The plate shall be in accordance with 3.35.2.

3.17.2 Fungus- and moisture-resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I with 1-percent copper 8-quinolinolate (by weight) or 7-percent salicylanilide based on the nonvolatile content of varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

The varnish shall be applied by spray, brush, or a combination of both, to give a minimum dry-film thickness of 1-mil to component or element surfaces. These surfaces shall be previously cleaned and prepared so that they are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.18 Blade. The blade assembly shall consist of a wear-resistant, high carbon, steel mold-board and replaceable cutting edges and shall have hydraulic sideshift, hydraulic tilt, and hydraulic blade float. The blade shall be the contractor's standard 12-foot-long blade, not less than 2 feet high and 0.75 inch thick.

3.19 Scarifier. The grader shall be equipped with a V-type scarifier. The scarifier shall be equipped with removable shanks having replaceable points. The scarifier shall have not less than nine shanks and a swath of not less than 46 inches.

3.19.1 Scarifier shank storage container. A storage container shall be of sufficient size to hold all scarifier shanks with replaceable points attached.

3.20 Front tow hook. A tow hook or other suitable means of attaching a tow cable shall be attached to the front of the grader.

3.21 Rear tow hitch. The grader shall be equipped with a rear tow hitch, including a pin and a positive pin retaining means.

3.22 Tools. Unless otherwise specified (see 6.2), any nonstandard tools required for equipment operation and operator maintenance of the grader shall be

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furnished with the grader. Any other specialized maintenance tools or equipment required to service the grader shall be provided.

3.22.1 Toolbox. A toolbox shall be of sufficient size to hold all tools required in 3.22. The toolbox shall have drain holes appropriately placed to prevent any moisture from accumulating.

3.23 Special requirements.

3.23.1 Blackout lighting. A blackout lighting system shall be furnished to meet the following requirements:

a. Each vehicle shall be equipped with the following blackout lights:

(1) One left front corner light conforming to MS51318.

(2) Two front corner marker lights conforming to MS35424-1.

(3) Two rear corner marker and stop taillights conforming to MS51330.

b. The blackout system shall be controlled by an interior switch, readily accessible to the driver, which shall prevent accidental disengagement of the blackout system from the blackout mode.

c. All regular exterior lights and horn, if electric, shall be made inoperable automatically when in the blackout mode.

d. The interior lighting shall have a variable brightness control switch readily accessible to the driver.

e. A separate wiring harness shall be provided for the blackout lights only if the required blackout lights cannot be powered using the standard wiring harness.

f. The blackout lights shall be oriented in the appropriate direction for roading of the grader. The lights shall be positioned at the following locations:

(1) Blackout driving lamp - mounted between the center and left-hand side of the vehicle, as far forward and as near the line of the operator's vision as practicable,at a height of approximately 42 inches above the road surface.

(2) Front blackout marker lamps - one on each side of thevertical centerline, as far apart as practicable and at the same level, between 15 and 72 inches above the road surface.

(3) Rear corner marker and stop taillights - one on eachside of the vertical centerline, as far apart as practicable and at the same level, between 15 and 72inches above the road surface.

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Changes to the locations outlined above will be subject to the approval of the contracting officer.

3.23.2 Winterized grader. When specified (see 6.2), a quantity of graders shall be winterized to permit starting and operating in arctic climates at ambient temperatures to -50 °F and storage at ambient temperatures to -60 °F. The contractor shall furnish one grader for examination and test within the timeframe specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce winterized graders that comply with the requirements of this specification.

3.23.3 Winterization kit. The kit shall consist of the components for preheating grader engine coolant and battery box; also insulated sound-suppressed ROPS cab; hot water personnel heater (equipped with one or more shut-off valve in the coolant lines to stop and vary the flow of coolant to the heater); defrosting system for cab windows; front and rear windshield wipers; radiator winterfront; insulated engine and radiator compartments; insulated battery box; and battery-finned connectors. Winterization components shall not affect the performance of the grader, and all other grader requirements shall be met as specified herein.

3.23.4 Heater and heat exchanger. The Eberspacher "Swingfire" heater, part number 26-9002, or equal, shall be used for preheat purposes. When specified (see 6.2), a quantity of heaters shall be furnished. The Eberspacher heat exchanger (water jacket), part no. 26-9013, or equal that is compatible with the Swingfire heater, shall be provided for heating the engine coolant. The heat exchanger shall be located near the engine compartment for easy accessibility for insertion of heater. The heater unit starting operation shall be provided by the grader's 24-volt electrical system, using a wiring harness with Bosh receptacle, part no. 135050012, or equal, compatible with the heater starter cable connector. After heater start and insertion into heat exchanger, preheating of the engine coolant shall be provided by thermosiphon action. The heater exhaust hot air shall be routed to the battery box for even, hot-air drenching of the battery box and then routed away from the battery box to the atmosphere. The hot airflow shall not be impeded by excessive back pressure or leakage. To retain the hot air, the system shall be insulated and heat shielded, as required.

3.23.5 Insulation. Insulation shall conform to MIL-I-14511, 0.50 inch. Adhesive for bonding the insulation shall retain its adhesive properties at ambient temperatures to -60 °F.

3.23.6 Kit performance. The winterization kit shall be of such design that the following criteria can be met when the grader has been cold soaked for 24 hours at a temperature of -50 °F:

a. Engine start within 5 minutes after preheating engine coolant and battery box for 90 minutes.

b. Within 1 hour after personnel heater start, the heater shall have raised and stabilized the cab temperature to +40 °F in the area occupied by the operator's torso when seated.

c. Within 1 hour after personnel heater start, the defrosting system shall have cleared sufficient glassed area for safe operation of the grader and

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visibility of the grader blade and work area normally seen when there is no frost on the windows.

d. All grader blade and scarifier controls shall be operational within an hour after engine start-up.

e. The transmission and brake controls shall be functional within an hour after engine start-up.

3.23.7 Kit installation. The winterized grader shall be capable of performing rough and fine grading, high bank cutting and sloping, grading in reverse, and scarifying.

3.23.8 Diagnostic connector assembly (DCA). Unless otherwise specified (see 6.2), the grader shall incorporate an easily accessible DCA in the operator's cab for interface with the Government's Simplified Test Equipment/Internal Combustion Engine (STE/ICE) specified in MIL-T-62314, appendix B. All requirements for the DCA shall be in accordance with TACOM Drawing 12258941. As a minimum, the DCA shall be capable of monitoring the functions defined in table II of report CR 82-588-003 (see 3.36) and the following: alternator/generator output voltage, alternator/generator field voltage, alternator/generator negative cable voltage drop, fuel solenoid voltage, starter voltage, starter current first peak and average starter circuit resistance, battery internal resistance, battery internal resistance change, fuel supply pressure and fuel filter restriction (Delta P). Additional test parameters to be monitored in either the DCA or transducer kit (TK) mode shall be as follows: fuel return pressure, fuel filter pressure drop, engine coolant temperature, starter solenoid voltage, battery resistance change, and transmission oil pressure. Test points identified to monitor functions in the TK mode shall be accessible without the removal of major assemblies and installed attachments. Those additional test parameters which are inaccessible for monitoring in the TK mode shall be monitored in the DCA mode. Fittings and adapters required for interface with the STE/ICE test set when monitoring in the TK mode shall be permanently installed on the grader to assure that adapters in the STE/ICE test set readily mate with TK mode test point locations. A separate wiring harness shall be provided for the DCA and shall include all wiring and necessary hardware to perform required capabilities. A fuel shut-off method shall be provided for conducting compression unbalance checks. The contractor shall also provide STE/ICE vehicle test cards in the format identified in TACOM Drawing 12258955 addressing both DCA and TK measurements and shall incorporate the test cards into the grader technical manuals (see 6.2.2).

3.24 Antivandalism protection. The grader shall be equipped with locks to secure the following:

a. Fuel tank filler cap.

b. Hydraulic oil tank filler cap.

c. Engine oil filler cap and dipstick.

d. Transmission oil filler cap and dipstick.

e. Radiator filler cap.

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f. Toolbox.

g. Battery box.

h. Cab doors.

When any of the above items are secured by lockable cab doors or lockable engine side panels, individual locks are not required. Keys for ignition, cab doors, cover panels, filler caps and tool battery boxes shall be interchangeable to the maximum extent possible. Two sets of keys shall be furnished with each grader, and each set shall be identical for every grader.

3.25 Lubrication. A lubrication means shall be provided for all moving parts requiring lubrication. Each grader shall be serviced and run-in with military lubricants conforming to MIL-L-2104, MIL-L-2105, MIL-L-46167, MIL-M-7866, MIL-G-10924, and MIL-G-23827, as applicable; however, initial servicing and run-in may be accomplished with lubricants standard to the contractor's commercial practice (except for the braking system) if compatible with above military lubricants.

3.25.1 Lubricants. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products List (QPL), the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.26 Performance.

3.26.1 Blade pull. The grader shall develop a blade pull of not less than 18,000 pounds at 2 $\pm$0.1 mph, at the specified operating weight, on a dry, level, concrete surface, with no more than a 15-percent slip factor.

3.26.2 Travel speed. The grader shall be capable of a forward travel speed of not less than 20 mph on level pavement.

3.26.3 Blade lift. The blade shall have a lift above ground of not less than 16 inches.

3.26.4 Blade cut. The blade shall have a cut below ground of not less than 10 inches.

3.26.5 Blade reach. The blade shall have a reach (right and left) of not less than 6 feet outside the rear tires.

3.26.6 Bank cutting angle. The blade shall have a right and left bank cutting angle of 90 degrees.

3.26.7 Circle turn. The blade shall be capable of traversing 360 degrees, without removing any machine components other than scarifier shanks, by using controls at the operator's station.

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3.26.8 Construction task. The grader shall be capable of performing rough and fine grading, ditching high bank cutting and sloping, grading in reverse, and scarifying.

3.26.9 Stability. The grader shall negotiate 35-percent side slopes and 60-percent longitudinal slopes without any wheel of the greater losing contact with the ground. All parts requiring lubrication shall be lubricated when the grader is operating under these conditions.

3.26.10 Fording. The grader shall start and operate during and after fording operations in fresh water to any depth up to 30 inches, including wave action, without damage and leakage of water into reservoirs containing lubricants.

3.27 Human factors. The grader shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control display integration (5.1), visual displays (5.2), audio displays (5.3), controls (5.4), labeling (5.5), anthropometry (5.6), design and maintainability (5.6), operational and maintenance ground/shipboard vehicles (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.28 Safety. All available standard options and accessories shall be included to provide protection of maintenance and operating personnel from rotating or reciprocating parts and form parts that are subject to high operational temperatures and so located as to be a hazard. The starting system shall be such that the transmission must be placed in neutral before the engine can be started, or the transmission must be placed in neutral before the grader can be set into motion. Exhaust discharges shall be directed so as not to endanger personnel or obstruct operator's view. All platforms and steps shall have antiskid surfaces. A decal shall be furnished and shall be located in he area of the articulation point. It shall warn of personnel clearance during frame articulation and placement of the steering frame lock for grader transport, as required, to provide for safe maintenance. Contractor's standard grab irons shall be provided. No asbestos material shall be used in the manufacture of this vehicle.

3.28.1 Noise limits. The noise produced by the grader shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.20. The provision of MIL-STD-1474, 4.3 and 4.4, shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform to MIL-STD-1474, 4.3, except the sign shall state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET." The sign shall be readable at the distance indicated on the sign.

3.29 Environmental requirements. Without the use of any external means, equipped with only standard and optional equipment, the engine shall start within 5 minutes. The grader shall operate under full-load conditions within 15 minutes after engine start in any ambient temperature from -25 to 120 °F. The grader shall be capable of being stored for extended periods in temperatures ranging from -65 to +160 °F without damage. Cold start aids must be integral with the grader, and no aids may be used which rely upon external support such as an

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electrical power source. Hand-held aerosol cans are not permitted. The grader shall be capable of operating in hot, humid, tropical environments and dry dusty environments.

3.30. Decontamination unit. The decontamination unit shall have a bracket assembly conforming to MS53052 shall be mounted by the contractor. The location of the bracket shall be such that it is accessible to personnel and provides clearance to remove and emplace a 5 gallon container conforming to MIL-C-1283.

3.31 Electromagnetic radiation. The grader shall comply with the electromagnetic radiation limits of SAE J551.

3.32 Transportability. The grader shall be capable of being transported by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment without damage or permanent deformation. The grader shall be equipped with tiedown or slinging provisions.

3.32.1 Highway and water transportability. The grader, with ROPS and FOPS, exhaust stack, and precleaner removed, shall be within all state highway permit combination. The grader shall be transportability by general break bulk ships, partial containerships, RORO ships, barge-carrier ships, and LACV-30 and larger vessels of the U.S. Army Lighterage Fleet.

3.32.2 Air transportability. The grader shall be capable of being transported in the C-130, C-141, and C-5A aircraft. Cargo envelopes and allowable axle weights for transport in the required aircraft are found in MIL-STD-1791. Disassembly to achieve air transportability configuration shall be accomplished in not more than two hours by 2 persons, using only common tools and available lifting devices. All removed parts shall be palletized for air transport with the grader. The operator's manual shall include instructions for component removal when required for transport (see 6.2.2).

3.32.3 Rail transportability. The reduced grader shall be rail transportable in CONUS and NATO countries without restrictions. The grader with the ROPS/FOPS cab, exhaust stack, and pre cleaner removed, shall have a dimensional profile within the Gabarit International de Chargement (GIC) in conformance with MTMCTEA Pamphlet 70-1, outline diagram when loaded on a 50-inch (127 cm) high rail car. The operational profile shall be within the Association of American Railroads (AAR) diagram when loaded on a 50-inch high rail car. The grader shall satisfactorily complete the rail impact test of MIL-STD-810, using standard gear flatcars.

3.32.4 Loading capability. Removal of ROPS cab and other components to reduce overall height shall not prevent grader from self-loading on military transport equipment under its own power.

3.32.5 Fuel capacity. When shipped, the grader shall be provided with a minimum of 25 percent of its fuel capacity in the tank. For air transport in military aircraft, the grader shall have no more than 50 percent of its fuel capacity in the tank. During water transport, the grader shall have no more than 25 percent of its fuel capacity in the tank.

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3.32.6 Steering frame lock. A steering frame lock in accordance with SAE J276 shall be provided.

3.33 Slinging and tiedown.

3.33.1 Slinging attachments. The grader shall be provided with four slinging provisions conforming to MIL-STD-209, class 1 or 3, type II. The provisions shall enable the grader to be lifted in the normal traveling or operating position. Provisions shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the provisions. Provisions shall be fastened to members which shall withstand stresses in the amount and direction of pull specified for the provisions without weld failure or permanent deformation of the grader or the provisions. The slinging attachments shall be labeled "LIFT HERE".

3.33.2 Tiedown provisions. The grader shall be provided with integral provisions to permit adequate tiedown of the grader to the floor or deck of the transportation medium such as a rail flatcar. The tiedown provisions shall conform to MIL-STD-209 class 2 or 3, type II and shall withstand stresses in the amount and direction of pull specified without weld failure or permanent deformation of the grader or the tiedown provision. Tiedown provisions may also be used as slinging attachments when such provisions meet the requirements specified in 3.31.1. Tiedown provisions shall be labeled "TIEDOWN" Complete freight cars and truck trailers shall be furnished in the operator's manual. The tiedown provisions on all applicable items shall be located to provide a symmentrical tiedown.

3.34 Treatment and painting. The portions of the grader normally painted internally and externally shall be cleaned, treated, and painted conforming to MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat shall be camouflage green 383, conforming to MIL-C-46168. When specified (see 6.2), the grader shall be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns, except the first article may be camouflage green 383.

3.34.1 Camouflage pattern data. When specified (see 6.2), the contractor shall provide camouflage pattern data.

3.35 Identification marking.

3.35.1 Identification plates and marking. Each grader shall be furnished with an identification plate showing the contractor's model number, NSN, USA registration number, serial number, weight, date of manufacture and date of delivery. The identification plate shall be permanently affixed to the grader in a readily accessible and conspicuous location. Plates shall conform to MIL-P-514 or contractor's standard plates, subject to the approval of the contracting officer. All attachments or components removed or disassembled for shipment shall be matchmarked for proper reassembly. The grader and each attachment shall be identified in accordance with the contractor's standard method.

3.35.2 Instruction plates. The grader shall be equipped with instruction plates, including warnings, cautions, tire pressure diagrams describing any special or important procedures to be followed in assembly, operating, or servicing the garden and procedures for the removal and replacement of ROPS and

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FOPS. Plates shall conform to MIL-P-514 or contractor's standard plates subject to the approval of the contracting officer. Each plate shall be attached by screws, bolts, or rivets in conspicuous, protected locations. Labeling of instruction plates shall comply with the labeling requirements of 3.27.

3.35.3 Vehicle weight classification sign kit. The contractor shall apply the vehicle weight classification numbers to each vehicle, using a vehicle weight classification number kit conforming to MIL-S-40626. The kit shall be located on the front of the vehicle in a location approved by the contracting officer. The contracting officer shall assign the classification numbers to be displayed on the vehicle. Instructions for changing the numbers based on the vehicle configuration, shall be provided in the operator's manual.

3.35.4 Shipping data plates. Shipping data plates for each grader shall be furnished and shall show the silhouette of the grader, indicating the location and capacity of lifting attachments, tiedown provisions, center of balance, and shipping weight. Plates shall conform to MIL-P-514 or contractor's standard plates, subject to the approval of the contracting officer. The plates shall be attached to the left side of the grader and shall show the location as viewed from the left side.

3.35.5 Case. The contractor shall furnish the fasteners and shall install the Government furnished case on the grader (see 3.36).

3.35.5.1 Mounting bracket. Unless otherwise specified (see 6.2), the Government furnished bracket for securing a M16 rifle shall be mounted in or near the operators compartment.

3.36 Government-furnished property. The following property in the quality indicated will be furnished by the Government (see 6.4).

Item No.	Description	Quality
1.	Camouflage paint pattern	As required.
2.	Cases complete with maintenance forms	As required.
3.	Mounting Bracket NSN 2540-01-223-0041	As required.
4.	Vehicle Weight Classification Sign Kit. NSN 9905-00-565-6267.	As required.
5.	Report CR 82-588-003 STE/ICE Design Guide for Vehicle Diagnostic Connector.	As required.

3.37 Basic issue items. Unless otherwise specified (see 6.2), each item on the basic-issue item list shall be furnished with the grader.

3.38 Vehicle classification. When specified (see 6.2), the contractor shall submit vehicle classification (see figure 1).

3.39 Workmanship. Defective components shall not be furnished. Parts and assemblies which have been repaired or modified to overcome deficiencies shall not be used. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Boltholes and rivet holes shall be accurately punched or drilled and shall have burrs removed. Rivet heads shall be full,

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neatly made, concentric with the rivit holes, and in full contact with the surface of the member. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the weld.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit he Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection. The first article grader shall be examined (see 4.3.1) and tested (see 4.3.2) to determine compliance with the requirements of this specification. Prior to the examination and testing of the grader, the following shall be performed:

a. The engine to be installed in the grader shall be tested in accordance with SAE J1349. Nonconformance to SAE J1349 shall constitute failure of this test.
b. Service the grader with the lubricants specified.
c. Limber-up the grader as prescribed by the contractor. No run-in (in addition to the contractor's commercial run-in procedures) shall be performed.

4.3.1 First article examination. Prior to first article test, the grader shall be examined for the defects marked "X" in column 1 of table I. Presence of one or more defects shall be cause for rejection.

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4.3.2 First article tests. Upon successful completion of the examination specified in 4.3.1, the first article model grader shall be subjected to the tests marked "X" in column 1 of table II. All first article tests shall be run using MIL-T-83133, grade JP-8 fuel. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Individual examination. After successful completion of the tests specified in 4.4.2, each grader shall be examined for defects marked "X" in column 2 of table I. Presence of one or more defects shall be cause for rejection.

4.4.2 Individual tests. Each grader shall be subject to the tests marked "X" in column 2 of table II. The production run-in test (see 4.5.2.9) shall be run on either VV-F-800, grade DF-2 or MIL-T-83133, grade JP-8 fuel. Failure of any test shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The grader shall be examined for the defects listed in table I.

TABLE I. Examination schedule.

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
		Major	
X	X	101. Grader type not as specified. All standard or optional equipment for grader not furnished	3.1
X	-	102. First article model not as specified.	3.2
X	X	103. Assembly incomplete or incorrect. Material not of same quality as used on first article grader	3.3
X	X	104. Materials not as specified.	3.1
X	X	105. Material deterioration prevention and control not as specified.	3.4.1

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TABLE I. Examination schedule. (Cont'd)

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	106. Dissimilar metals not as specified.	3.4.1.1
X	X	107. Identification of materials not as specified.	3.4.1.2
X	X	108. Recovered materials not as specified.	3.4.2
X	X	109. Cab not as specified. ROBS and FOPS not furnished. Personnel heater, defroster fans, windshield wipers and rearview mirrors not furnished.	3.5
X	X	110. Seat not as specified.	3.6
X	X	111. Horn not as specified.	3.7
X	X	112. Back-up alarm not as specified.	3.8
X	X	113. Articulation position indicator not furnished.	3.9
x	x	114. Diesel engine not as specified.	3.10
X	X	115. Engine side panels not as specified.	3.10.1
		116. Oil sampling valve not as specified.	3.10.2
X	X	117. Engine cooling system not as specified.	3.10.3
X	X	118. Tachometer drive not as specified.	3.10.4

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TABLE I. Examination schedule. (Cont'd)

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	119. Transmission and torque converter not as specified.	3.11, 3.11.1, 3.11.2
X	X	120. Tandem oscillation not as specified. Differential not as specified.	3.11.3
X	X	121. Fuel tank capacity and fuel tank draining provisions not as specified.	3.12
X	X	122. Steering not as specified. Braking system not as specified.	3.13, 3.13.1, 3.13.2
X	X	123. Air brake system not as specified.	3.13.2.1
X	X	124. Hydraulic system not as specified.	3.14
X	X	125. Wheels and tires not as specified.	3.15, 3.15.1, 3.15.2
X	X	126. Electrical system not as specified.	3.16
X	X	127. Lighting not as specified.	3.16.1
X	X	128. Alternator not as specified.	3.16.2
X	X	129. Instruments and indicators not as specified.	3.16.3
X	X	130. Batteries not as specified.	3.17
X	X	131. Slaving components not as specified.	3.17.1

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TABLE I. Examination schedule. (Cont'd)

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	132. Fungus- and moisture-resistance protection not as specified.	3.17.2
X	X	133. Blade not as specified.	3.18
X	X	134. Scarifier not as specified.	3.19
X	X	135. Scarifier shank storage container not as specified.	3.19.1
X	X	136. Front tow hook not as specified.	3.20
X	X	137. Rear tow hitch not as specified.	3.21
X	X	138. Tools (if applicable) not furnished.	3.22
X	X	139. Toolbox not as specified.	3.22.1
X	X	140. Blackout lights not as specified.	3.32.1
X	X	141. Winterized grader not as specified.	3.23.2
X	X	142. Winterization kit not as specified, when applicable.	3.23.2
X	X	143. Heater and heat exhanger not as specified, when applicable.	3.23.4
X	X	144. Insulation not as specified, when applicable.	3.23.5
X	X	145. DCA no as specified.	3.23.8

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TABLE I. Examination schedule. (Cont'd)

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	146. Antivandalism protection not as specified.	3.24
X	X	147. Lubrication and Lubricants not as	3.25, 3.25.1
X	X	148. Human factor characteristics not as specified.	3.27
X	X	149. Safety characteristics and articulation warning decal not as specified.	3.28
X	X	150. Sound-level warning plate not as specified.	3.28.1
X	X	151. Grader not furnished with equipment to enable grader to start and operate in temperatures rangin from -25 to +120 °F. Grader not furnished with equipment to enable grader to be stored for extended periods at temperatures ranging from -65 to 160 °F.	3.29
X	X	152. Decontamination unit not as specified.	3.30
X	X	153. Electromagnetic radiation limits not as specified.	3.31
X	X	154. Highway and water transportability not as specified.	3.32.1
X	X	155. Air transportability not as specified.	3.32.2

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TABLE I. Examination schedule. (Cont'd)

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	156. Rail transportability not as specified.	3.32.3
X	X	157. Loading capability not as specified.	3.32.4
X	X	158. Fuel capacity not as specified.	3.32.5
X	X	159. Steering frame lock not as specified.	3.32.6
X	X	160. Slinging attachments not as specified.	3.33.1
X	X	161. Tiedown provisions not as specified.	3.33.2
X	X	162. Treatment and painting not as specified. Paint color not as specified.	3.34
X	X	163. Camouflage pattern data not as specified.	3.34.1
X	X	164. Identification plate and marking not as specified.	3.35.1
X	X	165. Instruction plate not as specified.	3.35.2
X	X	166. Vehicle weight classification sign kit not furnished.	3.35.3
X	X	167. Shipping data plate not as specified.	3.35.4
X	X	168. Case not installed as specified.	3.35.5
X	X	169. Government-furnished containers not installed on grader.	3.36

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TABLE I. Examination schedule. (Cont'd)

First Article	Individual	Defect	Requirement Paragraph
1	2	3	4
X	X	170. Basic issue items (if specified) not furnished.	3.37
X	X	171. Vehicle classification not furnished.	3.38
X	X	172. Cracked or broken parts or assemblies. Permanent deformation of any part or assembly.	3.39

4.5.2 Tests. Tests shall be conducted in accordance with table II. Prior to conducting any operational tests, all STE/ICE diagnostic tests specified in 3.23.8 shall be measured and recorded.

TABLE II. Test schedule.

First article	Individual	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	Operating weight.	4.5.2.1	3.1
X	-	ROPS and FOPS.	4.5.2.2	3.5
X	-	Fuel tank capacity.	4.5.2.3	3.12
X	-	Steering and braking systems.	4.5.2.4	3.13.1 3.13.2
X	-	Scarifier swath width.	4.5.2.5	3.19
X	-	Cold tests, when applicable.	4.5.2.6	3.23.6
X	-	Installation test, when applicable.	4.5.2.7	3.23.7
X	-	DCA capability.	4.5.2.8	3.23.8

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TABLE II. Test schedule.

First article	Individual	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	DCA-STE/ICE interfaces.	4.5.2.8.1	3.23.8
X	X	Production run-in.	4.5.2.9	3.26
X	-	Blade pull.	4.5.2.10	3.26.1
X	-	Travel speed.	4.5.2.11	3.26.2
X	-	Blade lift.	4.5.2.12	3.26.3
X	-	Blade cut.	4.5.2.13	3.26.4
X	-	Blade reach.	4.5.2.14	3.26.5
X	-	Bank cutting angle.	4.5.2.15	3.26.6
X	-	Circle turn.	4.5.2.16	3.26.7
X	-	Construction task.	4.5.2.17	3.26.8
X	-	Stability.	4.5.2.18	3.26.9
X	-	Fording.	4.5.2.19	3.26.10
X	-	Noise limits.	4.5.2.20	3.28.1
X	-	Environmental requirements.	4.5.2.21	3.29
X	-	Rail impact test.	4.5.2.23	3.32.3
X	-	Slinging attachments.	4.5.2.24	3.33.1
X	-	Tiedown provisions.	4.5.2.25	3.33.2

4.5.2.1 Weight. The weight of the grader shall be measured with a scale calibrated within 6 months of the test. Inability to meet the requirements of 3.1 shall constitute failure of this test.

4.5.2.2 Rollover protective structure (ROPS) and falling objects protective structure (FOPS). The ROPS shall be tested in accordance with SAE J1040 and the FOPS tested in accordance with SAE J231 to prove conformance to 3.5. Non-conformance to 3.5 shall be cause for rejection. The removal and replacement of

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the ROPS and FOPS shall be tested to prove to conformance to 3.32. Permanent deformation or damage to any part of the ROPS and FOPS, or inability to meet the requirements of 3.32, shall constitute failure of this test.

4.5.2.3 Fuel tank capacity. The grader fuel tank capacity shall be measured during conduct of the construction task (see 4.5.2.17) to prove compliance with the fuel capacity requirements of 3.12. Nonconformance to 3.12 shall be cause for rejection.

4.5.2.4 Steering and braking system.

4.5.2.4.1 Steering system. The grader steering system shall be tested in accordance with SAE J53. Nonconformance to 3.13.1 shall constitute failure of this test.

4.5.2.4.2 Braking system. The grader braking system shall be tested in accordance with SAE J1152. Nonconformance to 3.13.2 shall constitute failure of this test.

4.5.2.5 Scarifier. The scarifier swath width shall be measured to prove conformance to 3.19. Nonconformance to 3.19 shall constitute failure of this test.

4.5.2.6 Cold tests. The grader shall be cold soaked in a temperature controlled chamber to a temperature of -50 ^{0}F and tested as follows to prove conformance to 3.23.6.

a. Preparation.

(1) Ensure that the grader has been serviced with arctic coolant, fuels,lubricants, and hydraulic oils.

(2) Service Swingfire heater with arctic gasoline.

(3) Place grader in temperature controlled chamber.

(4) Instrument the grader to record as a minimum the following temperatures:

(a) Exhaust, engine heater outlet.

(b) Air, battery box, inlet, outlet, and all aides.

(c) Air, operator's compartment, operator torso (seated positioned).

(d) Air, operator's compartment, floor, roof, sides (midway to top).

(e) Air, hot water heater outlet.

(f) Electrolyte, all batteries.

(g) Coolant, engine block inlet.

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(h) Coolant, engine block outlet.

(i) Oil, engine sump, and gallery.

b. Procedure.

(1) Subject grader to a stabilized temperature of -50 °F for 24 hours.

(2) Bring Swingfire heater into cold chamber. Connect heater to grader electrical power source and start heater. (After heater start, disconnect from power source.)

(3) Insert heater into heat exchanger and preheat engine coolant and battery for 90 minutes.

(4) Record temperatures continuously or at intervals not greater than 5 minutes.

(5) Remove heater from heat exchanger and shut off motor.

(6) Start engine from grader battery.

(7) Start personnel heater and defrost fans 30 minutes after engine start.

(8) One hour after heater start, record operator compartment air temperatures and check defrost.

(9) Operate grader steering, blade, and scarifier controls for 10 minutes within the dimensional limits possible in the test chamber.

(10) Engage and operate transmission and brake system as dimensional limits of test chamber permit.

Nonconformance to 3.23.6 shall constitute failure of this test.

4.5.2.7 Installation test. The purpose of this test is to ensure the proper fit, form, and function of the winterization kit and that no operational or safety deficiencies exist after installation of the kit on the grader for which it is intended. The grader shall be tested for 10 hours to prove conformance to 3.23.7. It shall include 5 hours for rough and fine grading, 3 hours for high bank cutting and sloping, 1 hour for grading in reverse, and 1 hour for scarifying.

4.5.2.8 DCA capability. The contractor will perform a 100 percent visual inspection of all DCA wiring harnesses prior to installation on the grader to ensure wires and connections are free of error. After installation on the grader, test the DCA harness (reference chapter 5 of design guide for DCA) on each vehicle with a DCA tester prior to interfacing with STE/ICE. Nonconformance to 3.23.8 shall constitute failure of the DCA.

4.5.2.8.1 DCA - STE/ICE interfaces. Test the DCA for all measurements specified in 3.23.8 using STE/ICE. After completion of all first article testing, the test shall be repeated and the results compared to the first test performed as specified in 4.5.2. Nonconformance to 3.23.8 or failure of any

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component or wiring in the DCA harness, which results in inability to obtain the measurements previously recorded with the STE/ICE (see 4.5.2), shall constitute failure of this test.

4.5.2.9 Production run-in. Start and operate each grader as follows: Operate the grader in all gears forward and reverse at maximum speeds commensurate with conditions for one-half hour. Stop and restart the engine not less than five times. Operate all blade and scarifier controls for 5-minute periods. Hold each hydraulic control in each actuation position until the hydraulic system reaches relief valve pressure at least five times. Operate the grader through not less than 10 full right and 10 full left turns using maximum wheel lean at the maximum speed commensurate with safe operation. Make not less than five abrupt panic stops from a speed not less than 20 mph using the service brake. Release and reapply the parking brake not less than five times. Turn on all lights for the duration of the run-in. Evidence of any deficiency, need for adjustment or maintenance, malfunction of any control, or evidence of leakage of fuel, lubricants, or hydraulic fluid shall be cause for rejection of the grader by the Government until corrected to the Government's satisfaction. The Government may require additional run-in after deficiencies are corrected. Any defect found during production run-in shall be prima facie evidence that graders previously accepted are similarly defective unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly defective.

4.5.2.10 Blade pull. The blade pull of the grader shall be measured to prove conformance to 3.26.1. Compliance with 3.26.1 shall be determined from pull tests conforming to SAE J872. Nonconformance to 3.26.1 shall constitute failure of this test.

4.5.2.11 Travel speed. The travel speed of the grader shall be measured to prove conformance to 3.26.2. Nonconformance to 3.26.2 shall constitute failure of this test.

4.5.2.12 Blade lift. The blade lift above ground shall be measured to prove conformance to 3.26.3. Nonconformance to 3.26.3 shall constitute failure of this test.

4.5.2.13 Blade cut. The blade cut below ground shall be measured to prove conformance to 3.26.4. Nonconformance to 3.26.4 shall constitute failure of this test.

4.5.2.14 Blade reach. The blade reach (right and left) outside the rear tires shall be measured to prove conformance to 3.26.5. Nonconformance to 3.26.5 shall constitute failure of this test.

4.5.2.15 Bank cutting angles. The blade bank cutting angle shall be measured to prove conformance to 3.26.6. Nonconformance to 3.26.6 shall constitute failure of this test.

4.5.2.16 Circle turn. The blade circle turn shall be measured to prove conformance to 3.26.7. Nonconformance to 3.26.7 shall constitute failure of this test.

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4.5.2.17 Construction task. The grader shall be tested as follows for a total of 115 hours to prove conformance to 3.26.8. Nonconformance to 3.26.8 or any malfunction or breakage of mechanical or electrical components shall constitute failure of this test.

a. Test for 105 hours the grader's ability to perform rough and fine grading operations (to include V-ditching).

b. Test for 5 hours the grader's ability to perform high bank cutting and sloping operation.

c. Test for 1 hour the grader's ability to grade in reverse.

d. In the most severe soil conditions available at the test site, test for 4 hours the grader's scarifying ability.

4.5.2.18 Stability. The stability of the grader shall be measured to prove conformance to 3.26.9. Nonconformance to 3.26.9 shall constitute failure of this test.

4.5.2.19 Fording. The fording ability of the grader shall be measured to prove conformance to 3.26.10. Nonconformance to 3.26.10 shall constitute failure of this test.

4.5.2.20 Noise level test. Noise levels shall be measured conforming to MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than twelve equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.21 Environmental conditions. The grader shall be tested as follows to prove conformance with 3.29:

a. Service the grader with applicable lubricants.

b. Place the grader in a temperature-controlled chamber.

c. Obtain a stabilized temperature of -25 °F in radiator top tank and battery electrolyte.

d. Using the cold-start aid furnished, start the grader engine.

e. Operate grader steering, blade, and scarifier controls for 10 minutes within the dimensional limits possible in the test chamber.

f. Engage and operate transmission and brake systems as the dimensional limits of the test chamber permit.

The above test shall be repeated at +120 °F. Nonconformance to 3.29 shall be cause for rejection.

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4.5.2.22 Electromagnetic radiation. The grader shall be tested in accordance with SAE J551. Measured levels in excess of recommended levels shall constitute failure of this test.

4.5.2.23 Rail impact test. The first article grader shall be tested conforming to MIL-STD-810 rail impact test to show conformance to transportability requirements of 3.32.3. Inability to meet requirements of 3.32.3, any permanent deformation of failure, or in operability after test, shall constitute a failure.

4.5.2.23.1 Failure criteria. Failure to meet the requirements of 3.32.3 and 4.5.2.23 rail impact test shall constitute failure of this test.

4.5.2.24 Slinging attachments. The slinging attachments shall be tested to prove conformance to 3.33.1. Nonconformance to 3.33.1 shall constitute failure of this test. In addition, weld failure or permanent deformation of the grader or slinging attachments shall constitute failure of this test. The test procedure shall be as follows:

a. Lift the grader and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the grader. Determine the force and angles of application for each attachment. Measure the clearance between each sling and the grader. The largest angle between the sling leg and the vertical should be approximately 45 degrees.

b. Restrain the grader by anchoring the mainframe and subject each slinging attachment to a force equal to the design loads specified in MIL-STD-209. This force shall be applied in the direction as determined in (a) above. Hold each load for a minimum of 90 seconds.

4.5.2.25 Tiedown provisions. The tiedown provisions shall be tested to prove conformance to 3.33.2. Apply the design loads as specified in MIL-STD-209 to each of the tiedown provisions. Nonconformance to 3.33.2 shall constitute failure of this test. In addition, weld failure or permanent deformation of the grader or tiedown provisions shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking of each grader for level A or B shall be examined to determine compliance with the quality assurance provisions of MIL-G-762. Level C preservation, packing, and marking shall be examined to determine compliance with the requirements specified in section 5 herein. Presence of one or more defects shall be cause for rejection.

5 PACKAGING.

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each complete grader shall be preserved conforming to level A preservation requirements of MIL-G-762.

5.1.2 Level B. Each complete grader shall be preserved conforming to level B preservation requirements of MIL-G-762.

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5.1.3 Level C. Each complete grader and all items issued with the grader shall be preserved and packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished: Items with critical surfaces required for fit or function should be provided protection by the use of preservative coating, volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrapping, pack compartmentalization, or boxing of the individual item. The cooling system shall be filled with a fresh solution of 50-percent water and 50-percent antifreeze, conforming to MIL-A-46153, and a tag indicating the temperature to which the cooling system can be subjected before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating, in detail, the preservation and packing materials to be removed and any other servicing required prior to placing the grader in operation.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each complete grader, preserved as specified in 5.1, shall be packed conforming to level A packing requirements of MIL-G-762.

5.2.2 Level C. Each complete grader, preserved as specified in 5.1, shall be packed as specified for level A except the strapping shall be finish A. Sealing of openings into engine housing and shrouding of the engines shall not be required except for overseas shipment.

5.3 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES.

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The grader covered by this specification is intended for use in military construction such as rough and fine grading, low- and high-bank sloping, flat and V-ditching, scarifying, and general construction and maintenance of roads.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Timeframe required for submission of the first article model (see 3.2).

d. When nonstandard tools are not required (see 3.22).

e. When winterized graders are required and number of graders to be furnished (see 3.23.2).

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f. Timeframe required for submission of winterized grader for examination and test (see 3.23.2).

g. When heaters are required and number of heaters to be furnished (see 3.23.4).

h. When a DCA is not required (see 3.23.8).

i. Color required when other than as specified (see 3.34).

j. When camouflage patterns will be provided (see 3.34.1 and 6.4).

k. When mounting of rifle bracket is not required (see 3.35.1).

l. When items on the basic issue items are not required (see 3.37).

m. Timeframe for completing and submitting the vehicle classification (see 3.38).

n. Level of preservation and packing required (see 5.1 and 5.2).

o. Any special marking (see 5.3).

p. Timeframe for completing and submitting Technical Information Package (TIP) (see 6.7).

q. When transportability report is required (see 6.8).

6.2.2 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

	Paragraph No.	DID Title	DID NO.
a.	3.34.1	Color photograph prints	DI-MISC-80192
b.	3.34.1	Camouflage line art data	DI-MISC-80176
c.	3.23.8	Commercial-off-the-shelf (COSTS)	DI-IMSS-80527

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be reached to ensure that only current, cleared DID's are cited on the DD Form 1423.

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.2.2.1 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.36.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of nonstandard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants, Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.6 Vehicle classification. The contracting officer should arrange to furnish a vehicle classification specification sheet to the contractor and request the return of reviewed sheet to the Government. The information thus obtained should be utilized to determine the weight classification numbers for the grader. The contracting officer should then request the contractor to apply the numbers to the grader. The use of a Government-furnished sign kit conforming to MIL-S-40626 should be utilized where applicable (see 3.35.3).

6.7 Technical information package (TIP). When required (see 6.2), the contractor shall complete and submit a TIP within the timeframe specified depicting the loader and its components exactly as it will be furnished. Notation shall be made of all components which have not been in field use for a minimum of 1 year and shall include identification of the component or part previously used. The TIP shall be submitted with all bids and proposals. Copies of the TIP are available from the Belvoir Research, Development and Engineering Center, ATTN: STRBE-FMT, Fort Belvoir, VA 22060-5606.

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6.8 Transportability reports. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report conforming to AR 70-47 and S.O.P. 70-38 for the grade 95 days prior to delivery of the first article grader.

6.9 Subject term (key word) listing.

Grader, articulated frame
Grader, diesel-engine
Grader, heavy duty
Grader, motorized
Grader, pneumatic-tired
Grader, road

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.11 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LPD's (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

(Certified by/date 9/12/89)

Custodian:
Army - ME

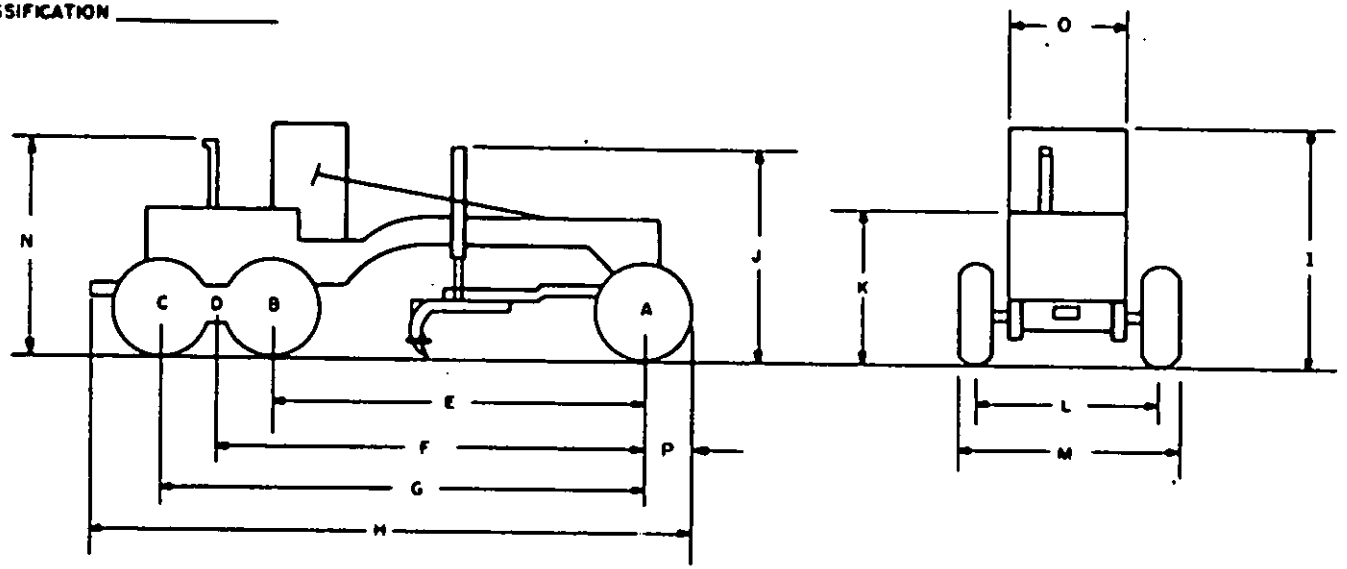
Review activities:
Army - AT
DLA - CS

Preparing activity:
Army - ME

Project 3805-A124

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NOMENCLATURE :
MODEL NUMBER :
MANUFACTURER:
FEDERAL STOCK NUMBER (FSN):
LOAD CLASSIFICATION _____



NUMBER OF AXLES
- A. WEIGHT ON FRONT AXLE AND WHEELS
- B. WEIGHT ON INTERMEDIATE WHEELS.
- C. WEIGHT ON REAR WHEELS.
- D. WEIGHT ON REAR AXLE.

TIRE SIZE.
TIRE PRESSURE.

TOTAL WEIGHT
- E. DISTANCE BETWEEN CENTERLINE OF FRONT AND INTERMEDIATE WHEELS.
- F. WHEELBASE.
- G. DISTANCE BETWEEN CENTERLINE OF FRONT AND REAR WHEELS
- H. OVERALL LENGTH
- I. OVERALL HEIGHT.
- J. HEIGHT TO TOP OF HYDRAULIC BLADE LIFT CYLINDER
- K. REDUCIBLE HEIGHT.
- L. TREAD WIDTH.
- M. WIDTH ACROSS TIRES.
- N. DISTANCE BETWEEN GROUND AND EXHAUST STACK.
- O. WIDTH OF CAB.
- P. DISTANCE OF CENTER FOR FRONT WHEEL.

FIGURE 1. Vehicle classification data sheet.

X-3746 A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-52996A(ME)	Grader, Heavy, Road, Motorized, DED, Commercial Construction Equipment (CCE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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